

The impact of a translating plunging jet on a pool of the same liquid

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An experimental study on the impact of a translating two-dimensional transient jet on an initially quiescent liquid pool is studied experimentally using high-speed cinematic visualization and particle image velocimetry methods. Six jet conditions (covering a range of jet thicknesses, velocities and inclination angles relative to vertical) are considered, with measurements performed over a range of horizontal translation speeds for each jet condition. For all conditions studied herein, the jet penetrates into the pool and forms two craters – one upstream and one downstream of the jet. Gravity acts to close these craters, which after a short time pinch off at intermediate depths, thereby entrapping cavities of air. The translation speed of the jet is found to have a dramatic effect on the cavity shapes, pinch-off depths and pinch-off times. A simple theory based on a potential flow and a hydrostatically driven collapse is used to model this flow, and the resulting jet tip trajectories and cavity shapes compare favourably with the experimental data.

Key words: air/sea interactions, bubble dynamics, gas/liquid flows

1. Introduction

The entrainment of air by a plunging liquid jet into a receiving pool of the same fluid is of particular interest for engineering applications within the geophysical, chemical and energy sciences due to its importance in aeration of the ocean by plunging breaking waves (Chanson & Lee 1997), its simplicity and efficiency as a gas/liquid reactor (Burgess & Molloy 1973), its occurrence behind weirs in open channel flows and as a mechanism for breaking down slugs of gas in two-phase pipe flows (see Kockx, Nieuwstadt & Oliemans 2005).

While nearly all of the previous studies have considered a steady stationary jet plunging into a stationary pool, for many practical applications the pool is moving at a constant speed and/or the jet is unsteady in time. Examples of jets plunging into moving streams include industrial casting processes, hydraulic effluent applications and the plunging bow waves of boats and ships when moving at a high speed. Unsteady jets are typical of wind-generated breaking waves on beaches or open water where the crest curls over and the resulting jet plunges into the water, creating a splash ahead of the jet impact point. In this case, the jet is unsteady in time and

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plunges into a pool of the same fluid that is moving relative to the translating jet. The current work documents and models the air entrainment process resulting from the transient impact of a translating planar jet, with the motivation of isolating the independent variables of jet thickness, impact velocity, translation velocity and impact angle.

Within steady non-moving plunging jet systems, several qualitatively different entrainment mechanisms have been observed which to first order depend upon the viscosity of the fluids, the impact velocity, the jet thickness, the irregularity of the jet surface and the ambient turbulence level in the receiving pool. Comprehensive reviews on the steady jet entrainment can be found in the article by Bin (1993) and the monograph by Chanson (1996), which summarize the details of what is known about these flows across a wide range of conditions. Since the current work is concerned with the translating transient impact problem, no further review of the steady non-translating case will be given here, except where it is directly relevant.

The literature on air entrainment due to unsteady and/or translating plunging jets is much less extensive than that for the steady non-translating case and presents a number of additional parameters to be considered. Such flows have been studied as a fundamental idealized problem to explore the mechanisms of air entrainment and also as an applied problem in the exploration of air entrainment by plunging breaking waves in the context of air/sea interaction. Given that the motivation for the majority of these studies comes from air/water flows, the Reynolds number is presumed to be large, and the effects of viscosity to be of secondary importance.

Focusing first on the condition of a non-translating transient impact, the simplest case is the impact of a finite volume of liquid on a free surface. When the impacting liquid is formed into a sphere, this corresponds to the case of a droplet hitting a quiescent pool, as reviewed in the article by Rein (1993) and explored in the studies of Pumphrey & Crum (1989), Prosperetti, Crum & Pumphrey (1989) and Oguz & Prosperetti (1990). The initial impact of the fluid generates an expanding hemispherical crater in the pool, typically rimmed by a splash. The growth of the crater is counteracted by the ambient hydrostatic pressure, which eventually reverses the crater wall motion and causes the cavity to collapse. Perhaps contrary to intuition, this geometry only entrains air over a narrow range of droplet diameter and velocity conditions. The regime of air entrainment is bounded by the reversal of the downwards surface motion at the bottom of the crater prior to the inward collapse of the crater walls, all of which is subtly modulated by the effects of surface tension. The result leads to a $We \propto Fr^{1/4}$ and $We \propto Fr^{1/5}$ scaling (where the Weber and Froude numbers are based on the diameter and speed of the droplet just before impact) for the upper and lower entrainment boundaries, respectively, as observed experimentally by Prosperetti *et al.* (1989) and later in numerical simulations by Oguz & Prosperetti (1990).

Changing the geometry of the impacting mass to an axisymmetric column forms the basis for studying air entrainment by short-duration jets, with an approximation of the drop case being recovered when the aspect ratio of the column is close to unity (*albeit* a non-spherical droplet). Kolaini, Roy & Crum (1991) and Kolaini *et al.* (1993) have experimentally considered the problem of a stationary short-duration water jet impacting on a quiescent pool of water. In these experiments, a vertically oriented tube of water was held in place above the pool, which was subsequently released by the sudden rupture of a rubber membrane, allowing a cylinder of water to impact with the pool. The emphasis of these studies was on the sound generated during the impact and the total captured volume of air and average void fraction. The motion of the air

cavity created during the impact was numerically simulated in a subsequent paper by Oguz, Prosperetti & Kolaini (1995). An elegant scaling argument was constructed on the basis of these results, using a potential flow Rankine body as the source of radial momentum to form the cavity, counteracted by a local time-dependent gravitational collapse. The results lead to a volume of entrained air proportional to $Fr^{1/3}$. Later work by Soh, Khoo & Yuen (2005) provided an integral analysis of the impact, predicting that the maximum depth of the cavity varies as $Fr^{1/3}$ to leading order, in agreement with the prior work. It should be mentioned that the dynamics of these cavities are similar to the dynamics of cavities created by the impact of solid objects on liquid-free surfaces, see, for example the recent article on cavities created by the impact of circular discs in Bergmann *et al.* (2009).

The above studies were further augmented by considering the case of an initially non-entraining plunging jet perturbed by a small surface undulation (see Ohl, Oguz & Prosperetti 2000; Zhu, Oguz & Prosperetti 2000; Kersten, Ohl & Prosperetti 2003), which has provided insight into the mechanisms of how air entrainment is established in highly disturbed low-velocity jets. These studies performed both experimental and numerical observations of the disturbance growth propagating along the jet as well as the subsurface flow generated by the ‘bump’. Upon contact of the disturbance with the receiving pool, an initial ‘crater’ was formed similar to that created by an impacting droplet. In contrast to the droplet case, however, the pre-existing subsurface flow encouraged the formation of a thinner secondary sheath or cavity of air extending from the bottom of the crater. It was this secondary cavity which would subsequently pinch off and form a submerged bubble, similar to the case created by a falling cylindrical mass of liquid (Oguz *et al.* 1995). Consequently, the scaling of the entrained air volume also follows a $Fr^{1/3}$ relationship. Tracking the energetics of the disturbance indicated that the momentum of the jet was being utilized to augment the air entrainment relative to an equivalent similarly sized droplet.

Fundamental work on translating jets has been reported by Chirichella *et al.* (2002), who documented the entrainment boundaries produced by a laminar axisymmetric steady jet translated over a quiescent pool. They found that the inception boundary was given by $Fr_t = V_t^2/(gD) \sim 1$, where V_t is the translation velocity of the jet, D is the diameter of the jet at impact and g is the acceleration constant due to gravity. The mechanism of the entrainment was a destabilization of the cusp-like free surface in the wake of the impingement region, driven by the coherent structures produced by the Kelvin–Helmholtz instability of the submerged jet.

The majority of the available studies on air entrainment due to a translating plunging jet has been examined by studying plunging breaking water waves directly. This jet is, of course, also unsteady. Melville (1996) has given a review of the effect of breaking waves on the air–sea interaction which addresses the literature on the air entrainment associated with the unsteady breakers. A rather striking sequence of photographs illustrating the air entrainment process in plunging breakers appears in the doctoral thesis of Lamarre (1993) and can be found in Melville’s review. The photographs show that the jet approaches the water surface (which is moving towards the wave crest at the phase speed of the wave) at an angle considerably less than 90° . The subsequent impact creates a splash that projects out ahead of the initial jet impact point. Two zones of entrained air bubbles can be seen: one at the site of the initial jet impact point and one at the site of the splash impact. This process has been observed by other investigators as well (see for example Bonmarin, Rochefort & Bourguel 1989). Recent unsteady numerical simulations of three-dimensional (3D) plunging breakers (Lubin *et al.* 2006) and fully resolved two-dimensional (2D) breaking events (Iafrati

2009) provide details of the impact–splash–impact process. Specifically, the work of Iafrati (2009) details the air entrainment and degassing processes as a function of initial wave steepness (which alters the jet impact geometry and kinematics), finding that the initial air entrapped by the jet scales with the initial wave steepness to the seventh power.

Those measuring the bubbles generated by breakers in wave tanks or the open ocean are generally interested in the void fraction, bubble size distributions and radiated noise (see, for example Kolaini & Crum 1994; Loewen & Melville 1994; Deane & Stokes 2002). These data are primarily correlated with the characteristics of the wave, for example the wave frequency and ‘breaking intensity’, rather than the characteristics of the jet itself and the motion of the free surface upon which the jet impacts. Thus, the fundamental processes that occur around the jet and determine the air entrainment, the bubble size distributions and the subsequent bubble motions are not studied in detail. These aspects of the breaker flows cannot be determined from studies of steady stationary jets as described above.

The bubble entrainment by the plunging breaker at the bow of a fast moving ship has been studied experimentally with a model flow in an article by Waniewski, Brennen & Raichlen (2001). The flow is actually a shallow-water flow with a surface-piercing wedge representing the ship bow. The water rises up the side of the wedge and peels off into the air, eventually plunging into the flow far from the wedge. Conductivity probes were used to measure the shape of the water surface and the occurrence of entrained air bubbles due to the plunging. Preliminary results point to a relationship between the roughness length scale and velocity of the jet surface and the frequency of bubble production in the underlying flow.

In the present research, the development of craters and the entrapment of air as the leading edge of a two-dimensional jet that hits the surface of a quiescent pool of liquid are examined experimentally. Although ostensibly motivated by applications of plunging breaking waves, the current work only covers a narrow range of impact angles and velocities due to the limitations of the experimental facility. While these test cases represent a narrow spectrum of plunging breaking wave impact conditions, the results are, of course, relevant to other artificial plunging jet systems and provide novel insight into the air entrainment process. Six jet conditions with various combinations of jet thickness, impact velocity, translation speeds and angle relative to vertical were studied. The experimental details of this research are described in §2. The results of these experiments are presented and discussed in light of a simple model in §3. Finally, the conclusions of this work are given in §4.

2. Experimental details

2.1. Facilities

The experiments were performed in a towing tank that is 7.32 m long, 0.76 m wide and 0.91 m deep. The tank walls and bottom consist of glass panels supported by a steel frame. The tank includes a water filtration system in which water is extracted via a surface skimmer placed at one end of the tank. From the skimmer, the water passes through a diatomaceous earth filter and then back into the tank at the end opposite to the skimmer.

An instrument carriage is supported by bearings that ride on two round rails mounted on the tops of the long side walls of the tank. The carriage is propelled by wire ropes, which are in turn driven by a servo motor. The carriage speed is controlled by the servo motor drive via an optical encoder feedback system and was checked by

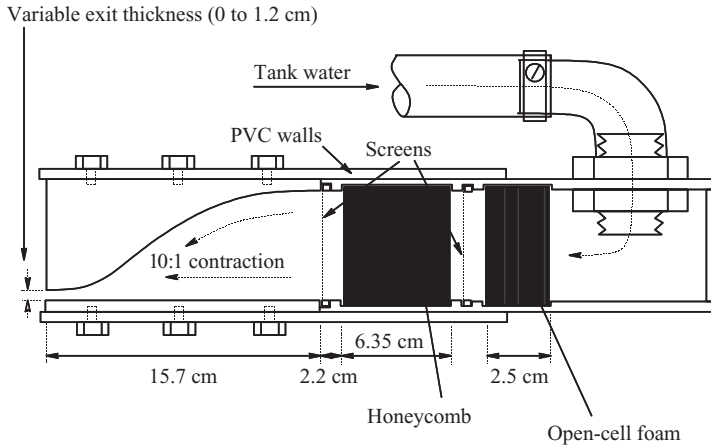


FIGURE 1. Schematic of the nozzle.

measuring the time of travel of the carriage over a fixed distance. It was found that the carriage speed varied by less than 0.01 % from run to run.

A schematic drawing of the jet nozzle is shown in figure 1. A nozzle design with one flat wall was chosen to allow the jet exit to be placed as close as possible to the water surface. The nozzle exit is 45.7 cm wide and its thickness can be varied from 0 to 1.2 cm. The nozzle includes a system of screens and a honeycomb to condition the flow before it enters a contraction. The contraction has an area ratio of 10 : 1 when the exit thickness is set to 0.5 cm. The shape of the contraction was designed to have a favourable pressure gradient throughout its length by using potential flow theory (see Gomez-Ledesma 2004). Water was supplied by a centrifugal pump whose suction line was connected to a drain in the bottom of the towing tank. The pump was mounted to the laboratory floor and its output was connected to the nozzle via two flexible hoses.

A jet cutter system was used to create the transient impact of a two-dimensional jet on a calm water surface starting with the above-described steady jet nozzle-pump system. The jet cutter is shown schematically in figure 2(a) and 2(b). The nozzle and the cutter were mounted on the instrument carriage. The cutter consisted of a stainless steel plate that was pivoted around its rotation axis by a pulley system and two pneumatic pistons. When the plate is in its starting position (figure 2a), the jet hits the back side of the plate and the water is deflected away from the impact site. When the cutter is activated, it accelerates rapidly and cuts through the water sheet, creating a leading edge that moves over the rotated cutter and impacts the water surface (figure 2b). In order keep the water surface calm before impact, a fixed dividing wall is placed in the tank as shown in the figure. A microswitch triggers a solenoid valve to initiate the cutter motion. The microswitch is positioned so that the jet cutter is triggered as the carriage approaches the barrier and the deflected jet hits the water in front of (to the right of) the barrier (figure 2a), while the jet impact occurs behind (to the left of) the barrier (figure 2b). Photographs of the leading edge of the jet before impact indicate that it is rounded with a diameter larger than that of the jet thickness immediately upstream. This leading-edge bulge is probably the result of the short but finite time that it takes the cutter to move through the jet and surface tension forces acting between the time when the jet leading edge is created and the time of impact. Although difficult to determine with precision, the diameter

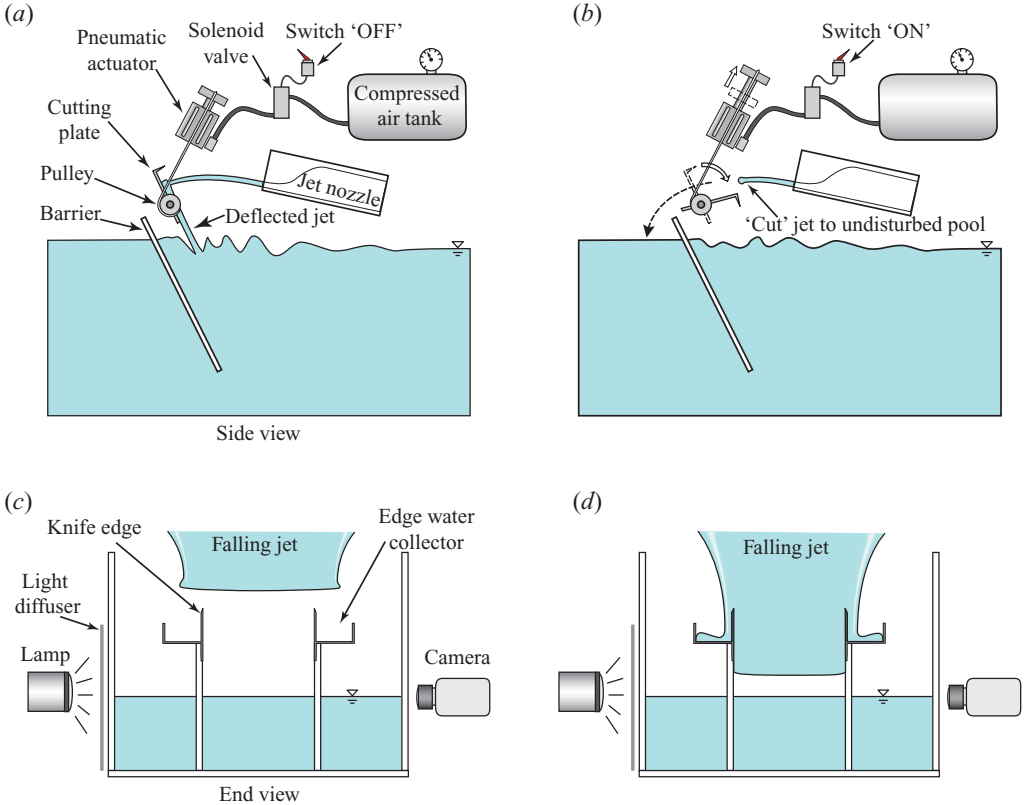


FIGURE 2. (Colour online available at journals.cambridge.org/FLM) (a) and (b) Side views of the cutter used to create a transient impact of the plunging jet. (c) and (d) End views of the apparatus used to remove the thick right and left edges of the jet.

of this leading-edge bulge was estimated from close-up images to be no more than five times the jet thickness for any of the experimental conditions used herein.

As the jet comes out of the nozzle exit, surface tension causes the edges of the jet to contract, thus reducing the jet width and creating thick edges as the jet approaches the water surface. To eliminate this effect, a structure consisting of two vertical Plexiglas plates and metal troughs was placed in the tank, see figure 2(c) and 2(d). The plates isolate the central part of the jet and the troughs capture the sides of the jet that are removed, keeping them from hitting the water and obscuring camera's view of the central section of the tank.

2.2. Cavity shape measurements

A Vision Research, Phantom V4.0 high-speed digital movie camera was used to visualize the jet impact at $1000 \text{ frames s}^{-1}$ with a resolution of 512×512 pixels. The camera was set to look through the side tank walls, with the jet and the receiving pool back-lit by a continuous light source positioned on the opposite side of the tank, as shown in figure 2(c) and 2(d). A thin translucent plastic sheet was fixed to the tank wall on the side of the light source. This sheet acted as a diffuser to provide a more uniform lighting. Both the camera and the lights were fixed to the carriage, keeping the image reference frame stationary relative to the water jet during its horizontal translation. The camera was controlled by a laptop computer that was also attached

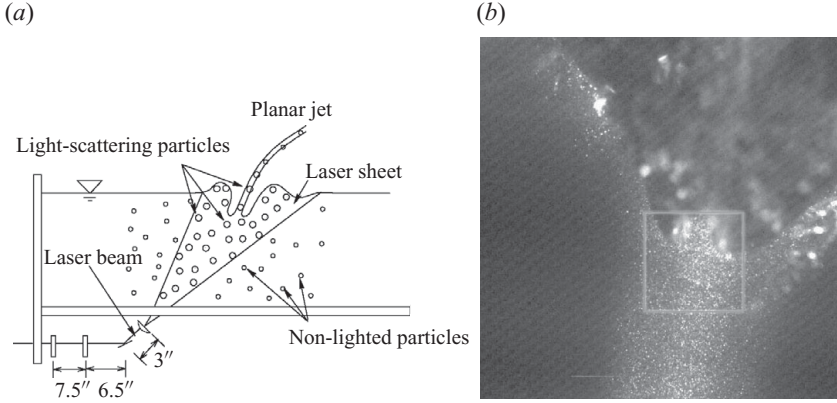


FIGURE 3. PIV measurements. (a) Lateral view of the PIV setup. (b) PIV image for condition III. $V_j = 0.3 \text{ m s}^{-1}$. The image shows the PIV particles and the two underwater air craters. The camera is translating with the jet from right to left. The rectangle ($17 \text{ mm} \times 17 \text{ mm}$) is the region where the PIV processing was applied.

to the carriage. The camera had a simultaneous view both above and below the tank mean water line to allow for the determination of the jet properties prior to impact, in addition to tracking the evolution of the subsurface cavity shape. Before each set of runs, images of a ruler placed in the tank on the Plexiglas inner wall nearest to the camera were taken to allow for the physical scaling of the different jet features measured in the movies.

2.3. Subsurface flow-field measurements

For several of the experimental conditions, the subsurface flow field in the vicinity of the nose of the craters was measured using a cinematic particle image velocimetry (PIV) technique. In order to take the PIV measurements, the tank and jet water were seeded with $25 \text{ }\mu\text{m}$ spherical hollow glass particles and the flow was illuminated with 12 mJ light pulses generated by a high-repetition-rate Nd:YAG laser (Clark MXR, model ORC-1000). The PIV images were recorded with the Phantom V4.0 high-speed digital movie camera. The laser beam was expanded into a light sheet by using a cylindrical lens (225 mm focal length), while a combination of two additional cylindrical lenses (75 and 100 mm focal lengths) were used to give a thickness of approximately 1 mm to the light sheet in the measurement area. A set of mirrors were used to project the light sheet upwards through the glass bottom of the tank and position it in the mid plane between the two vertical Plexiglas walls of the jet edge cutoff system (figure 2c,d) as shown in figure 3(a). The light sheet was stationary with respect to the tank and positioned along the tank length such that the jet plunged into the region illuminated by the laser. The camera was attached to the carriage with an orientation that allowed for an upwards viewing angle (about 10° from horizontal) through the glass vertical side walls of the tank. Calibration was achieved by photographing a calibration grid placed in the plane of the light sheet.

The camera was set to operate at $1000 \text{ frames s}^{-1}$, and the exposure time of each frame was nearly 1.0 ms . During this exposure time, the background light intensity was insufficient to register on the image sensor, rendering the effective exposure time to be the duration of the laser pulses ($\approx 500 \text{ ns}$). The camera was programmed to send a 5-volt square-wave output signal at the beginning of each frame, which controlled the timing of a pulse generator for the laser to give frame-straddled images. This

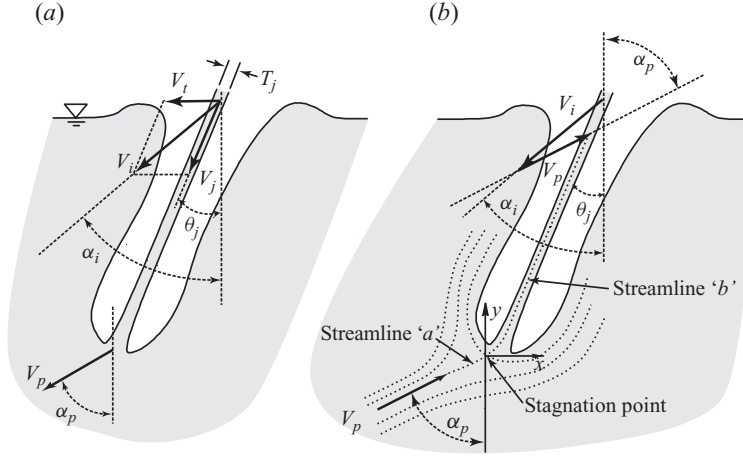


FIGURE 4. Schematic of the underwater jet and crater with various parameters shown. The velocities are shown (a) in a reference frame fixed with respect to the water tank and (b) in a reference frame fixed with respect to the jet tip.

resulted in a pair of distinct PIV images with a temporal spacing of 0.40 ms acquired at a rate of 500 Hz.

Velocity fields were obtained from each pair of PIV images using cross-correlation techniques in DaVis 6.0 software. Processing was done in four passes with decreasingly smaller window sizes from 32×32 to 8×8 pixels and an overlap of 50 %. The areas of the frames that were contaminated by light reflections coming from the underwater crater walls were masked out from the processed vector fields. Figure 3(b) shows one of the PIV frames and the region where the PIV processing was applied.

2.4. Experimental conditions

The angle of incidence (θ_j , see figure 4a for definitions), jet speed (V_j) and jet thickness (T_j) at the point of impact of the jet on the undisturbed free surface of the tank are determined by the orientation of the nozzle, the vertical distance between the nozzle outlet and the free surface, the thickness of the nozzle opening and the water flow rate. The speed and angle of incidence of the leading edge of the planar sheet at impact were measured by tracking the lowest point on the leading edge over a number of frames in high-speed movies taken with the camera positioned just above the water surface and the carriage held stationary. The jet thickness at impact was calculated from the conservation of mass, $T_j = Q/(V_j L_s)$, where L_s is the span of the nozzle opening and Q is the nozzle flow rate which was determined by capturing and measuring the volume of water produced by the nozzle over a fixed measured time. The nozzle flow rate was set by controlling the speed of the centrifugal pump.

Six jet conditions were chosen for the present experiments and the corresponding average values and standard deviations (from several measurements at each jet condition) of V_j , T_j and θ_j are shown in table 1 along with the corresponding Froude number ($Fr_j = V_j/\sqrt{gT_j}$), Reynolds number ($Re_j = V_j T_j/\nu$) and Weber number ($We_j = \rho V_j^2 T_j/\sigma$), where g is the acceleration of gravity and ν , σ and ρ are the kinematic viscosity, surface tension and density of water, respectively. At each jet condition, measurements of the cavity profile history were taken at towing speeds of $V_t = 0.0, 0.15, 0.30, 0.45$ and 0.60 m s^{-1} .

Case	No. of tests	Mean $\pm$ standard deviation			Fr_j	Re_j	We_j
		V_j (m s ⁻¹)	T_j (cm)	θ_j (deg.)			
I	3	2.59 ± 0.05	0.38 ± 0.02	32.5 ± 1.9	13.6	8782	3.6
II	5	3.48 ± 0.11	0.28 ± 0.02	45.2 ± 1.1	21.0	8562	4.6
III	3	3.04 ± 0.01	0.36 ± 0.01	17.4 ± 1.6	16.2	9537	4.5
IV	2	3.32 ± 0.07	0.29 ± 0.02	7.1 ± 0.2	19.7	8460	4.4
V	3	2.12 ± 0.09	0.35 ± 0.02	29.5 ± 1.2	11.4	6520	2.1
VI	3	2.22 ± 0.07	0.50 ± 0.04	23.8 ± 1.3	10.0	9754	3.4

TABLE 1. Mean and standard deviation of the jet impact parameters for the six jet conditions used in the present experiments. The jet speed (V_j), the jet thickness (T_j) and the angle of the jet relative to vertical (θ_j) were determined at the level of the undisturbed water surface in the tank. See figure 4(a) for definitions. Also given in the table are the corresponding Froude numbers ($Fr_j = V_j / \sqrt{gT_j}$), Reynolds numbers ($Re_j = V_j T_j / \nu$) and Weber numbers ($We_j = \rho V_j^2 T_j / \sigma$), where g is the acceleration of gravity and ν , σ and ρ are the kinematic viscosity, surface tension and density of water, respectively.

3. Results and discussion

3.1. Observations of the cavity evolution and the flow field near the jet tip

To illustrate the main features of the free-surface motion typically seen in the experiments, images at four times, $t = 40, 80, 121$ and 160 ms ($t = 0$ is the time of initial impact, with the camera translating from right to left with the jet), during an impact event for jet condition VI with $V_i = 0.15$ m s⁻¹, are shown in figure 5(a–d), respectively. In this study, the region where the free-falling jet contacts the pool is generally referred to as the ‘jet tip’ or the nose of the craters. The details of how this location is quantified is given in the discussion on the subsurface flow. Following the initial impact, the jet continues to penetrate the water at an angle similar to the jet impact angle and the depth of the craters increases. The angle of the jet does deviate slightly from its initial impact angle due to the continued acceleration of the jet as it falls into the cavity, but this variation is typically smaller than the uncertainty in the measurement itself. During the early stage of development (as shown in figure 5b), the crater interface is observed to move away from the centreline of the jet for all depths and takes a characteristic inverted bell-like shape. After a sufficient time, the outwards crater wall motion at any fixed depth ceases, and thereafter the crater wall at this depth moves towards the jet. Eventually, the upstream (left) and downstream (right) crater walls touch the underwater jet at separate locations, producing a pinch-off of the air cavities (figure 5c). (Here and in the following, the nomenclature ‘upstream’ and ‘downstream’ is taken to be the motion of the oncoming quiescent pool viewed from the reference frame translating with the jet nozzle.) Typically, the pinch-off occurs first on the upstream side of the jet. The subsequent collapse of the cavity produces a cloud of bubbles that is entrained downstream in the form of a turbulent multiphase flow (figure 5d) that eventually rises under the influence of buoyancy. The above-described behaviour is qualitatively similar for all jet conditions.

To show the differences in the jet impact in the six jet conditions, one image for each jet condition for cases with no translation velocity ($V_i = 0.0$) is shown in figure 6. All of the images were recorded at $t = 100$ ms after initial impact. The wide variation in the depth of the tip of the craters is primarily due to the wide range of jet speeds, V_j , which vary from 2.12 m s⁻¹ in jet condition V to 3.48 m s⁻¹ in jet condition II. All the jets are tilted clockwise to various degrees (ranging from 7.1° to 45.2° from

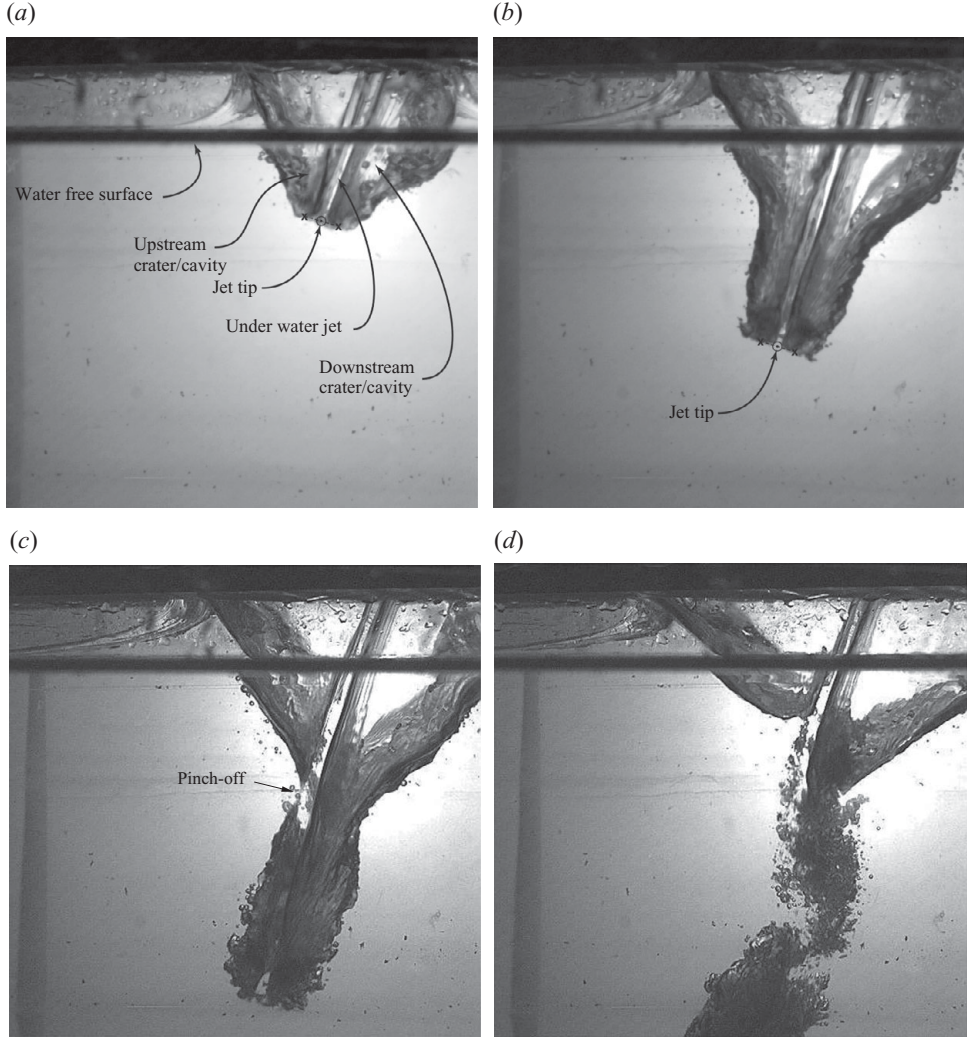


FIGURE 5. Photographs of the jet and crater for jet condition VI and a translation speed of $V_t = 0.15 \text{ ms}^{-1}$. The movie was taken in the reference frame of the jet nozzle and in this frame the undisturbed water in the tank moves from left to right with speed V_t . (a) $t = 40 \text{ ms}$, (b) $t = 80 \text{ ms}$, (c) $t = 121 \text{ ms}$ and (d) $t = 160 \text{ ms}$, where $t = 0$ is the moment of jet impact. The horizontal field of view of the images is 16.51 cm.

vertical at initial impact, see table 1), producing a horizontal component of velocity from right to left, which is also the direction of the carriage translation when $V_t \neq 0$. As can be seen from the images, even with $V_t = 0$, the craters created upstream of the jet are quite different from those created downstream of the jet.

Images from high-speed movies at the moment of the first pinch-off ($t = t_{po}$) for condition V at translation speeds of 0.15, 0.30, 0.45 and 0.60 ms^{-1} are shown in figure 7(a–d), respectively. In all the cases, the first pinch-off event occurs when the wall of the upstream (left) crater touches the upstream surface of the jet at a point above the jet tip. The jet translation induces a severe asymmetry between the upstream and downstream air craters. As V_t is increased, the downstream crater becomes wider and the upstream crater becomes thinner. Also, the outer boundary

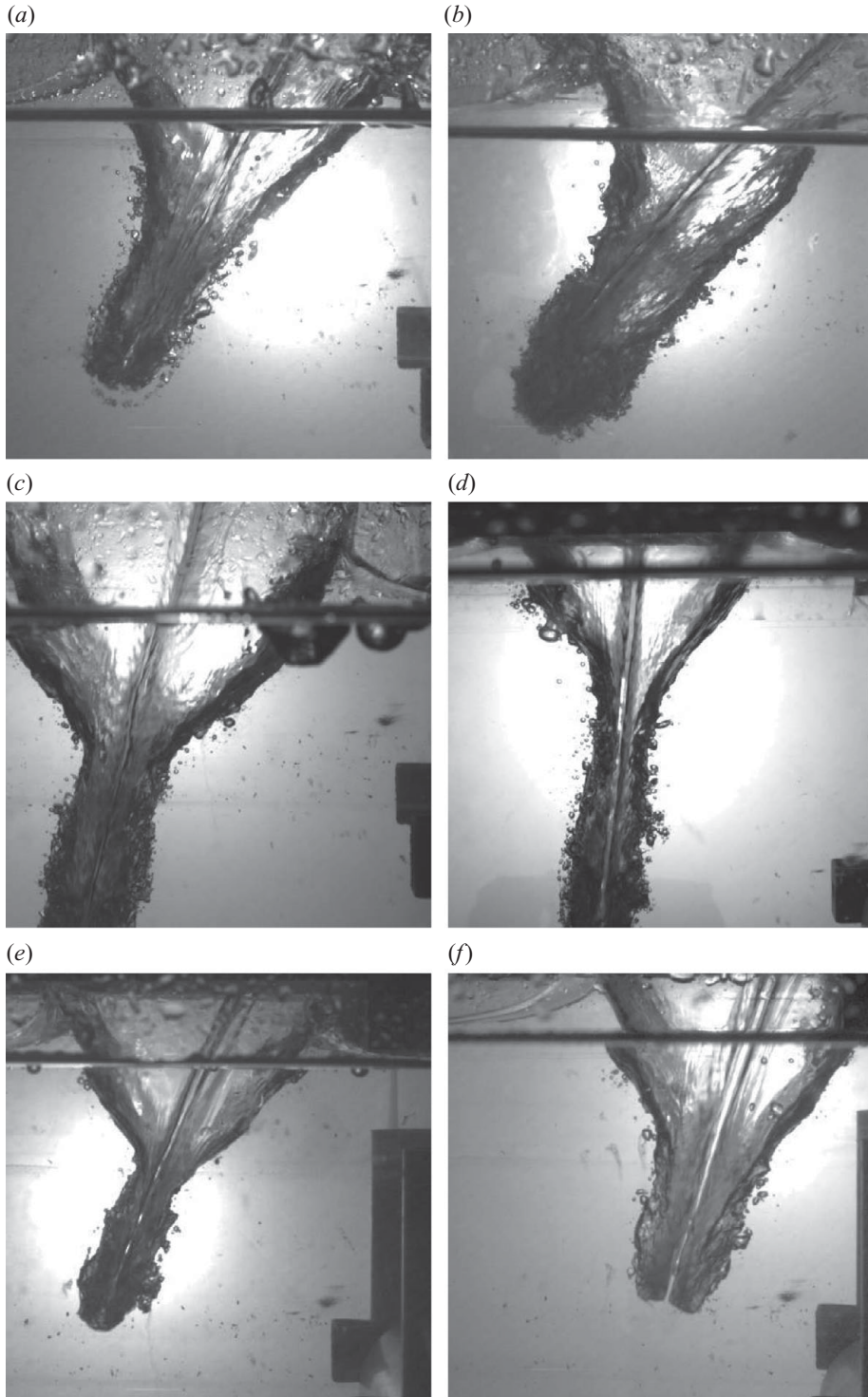


FIGURE 6. Photographs of the jet, upstream (left) crater and downstream (right) crater for all jet conditions at a time of 100 ms after initial impact. In all the cases, $V_i = 0.0$. (a) Jet condition I, (b) II, (c) III, (d) IV, (e) V (f) VI. The horizontal field of view of the images is 15.4 cm. Details of the six jet conditions can be found in table 1.

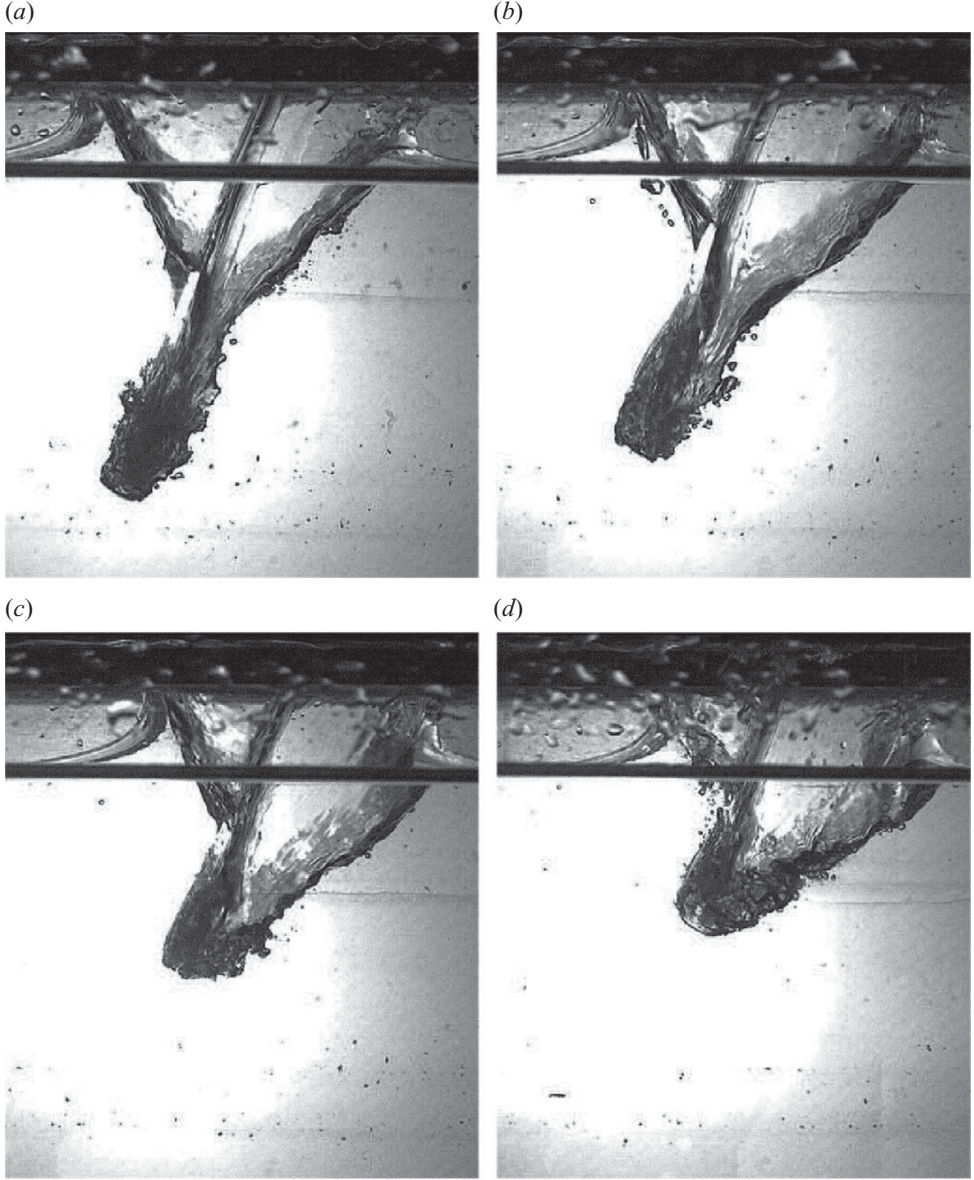


FIGURE 7. Photographs of the jet and crater at pinch-off of the forward (left) crater ($t = t_{po}$) for jet condition V and four jet translation speeds: (a) $V_t = 0.15 \text{ m s}^{-1}$ ($t = 106 \text{ ms}$); (b) $V_t = 0.30 \text{ m s}^{-1}$ ($t = 95 \text{ ms}$); (c) $V_t = 0.45 \text{ m s}^{-1}$ ($t = 71 \text{ ms}$); and (d) $V_t = 0.60 \text{ m s}^{-1}$ ($t = 57 \text{ ms}$). The movies were taken in the reference frame of the jet nozzle and in this frame the undisturbed water in the tank moves from left to right with the speed V_t . The horizontal field of view of the images is 17.43 cm.

of the downstream crater becomes more inclined towards the horizontal, while the outer boundary of the upstream crater becomes more vertical. Finally, the pinch-off time and depth decrease with increasing V_t .

PIV measurements of the flow near the underwater jet tip were performed for jet condition III at five different jet translation velocities: $V_t = 0.0, 0.15, 0.30, 0.45$ and

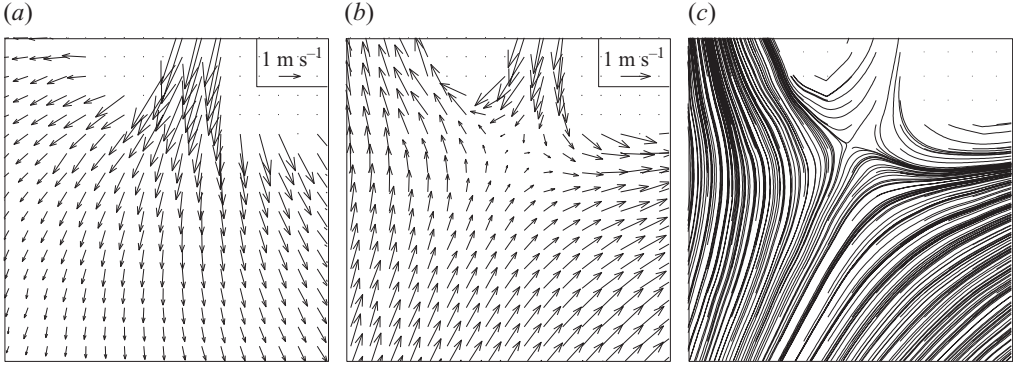


FIGURE 8. Velocity field for jet condition III and $V_t = 0.30 \text{ m s}^{-1}$ (a) in a reference frame moving with the jet carriage and (b) moving with the jet tip. The streamlines corresponding to the velocity field in (b) are plotted in (c). The flow field is shown over an area of $17 \text{ mm} \times 17 \text{ mm}$.

0.6 m s^{-1} . The PIV movies were used to determine both the velocity field in the water by using correlation techniques as described in §2 and the velocity of the jet tip by tracking the jet tip from frame to frame. (The velocity of the jet tip was also measured from the white light movies for all conditions, see below.) Figures 8(a) and 8(b) show the vector velocity field around the jet tip in a reference frame fixed with respect to the carriage (as measured) and in a reference frame fixed with respect to the jet tip, respectively for $V_t = 0.3 \text{ m s}^{-1}$. The streamline pattern corresponding to the velocity field given in figure 8(b) is shown in figure 8(c). These results are typical of the other four translation speeds and show that, as one would expect, in a reference frame moving with the jet tip, the underwater jet sees a uniform stream from the pool that moves upwards towards the plunging stream. The free stream of fluid from the pool curves closely around the air craters, except for one streamline that finds the jet water at a stagnation point. Two other streamlines branch off from the stagnation point, which are the dividing lines between the jet water and the incoming tank water stream. Flow fields for the other translation speeds for jet condition III show that as the jet horizontal translating speed is increased, the stagnation point moves slightly upwards and to the left with respect to the jet tip. The qualitative flow-field information near the underwater jet tip obtained from the PIV measurements is the starting point of the theoretical discussion presented below.

3.2. The stagnation point velocity

3.2.1. Measurements

The underwater jet tip speed relative to the laboratory reference frame (V_p) and angle (α_p) between V_p and the vertical (see figure 4a) were measured for all experimental conditions. The jet tip location at each instant would most logically be taken as the location of the stagnation point in the moving reference frame defined in the PIV measurements described above. However, PIV measurements were only available for a few cases. Thus, the jet tip location was determined from the high-speed movies of the jet impact (as shown, for example in figure 5) by clicking on a point at the bottom of the upstream and downstream craters, defining the jet tip to be at the mid-width of the jet. To get the jet tip velocity, its location was determined in this manner for five movie frames equally spaced in time intervals of 15 ms. It was found that the frame-to-frame displacements were nearly constant for each experimental

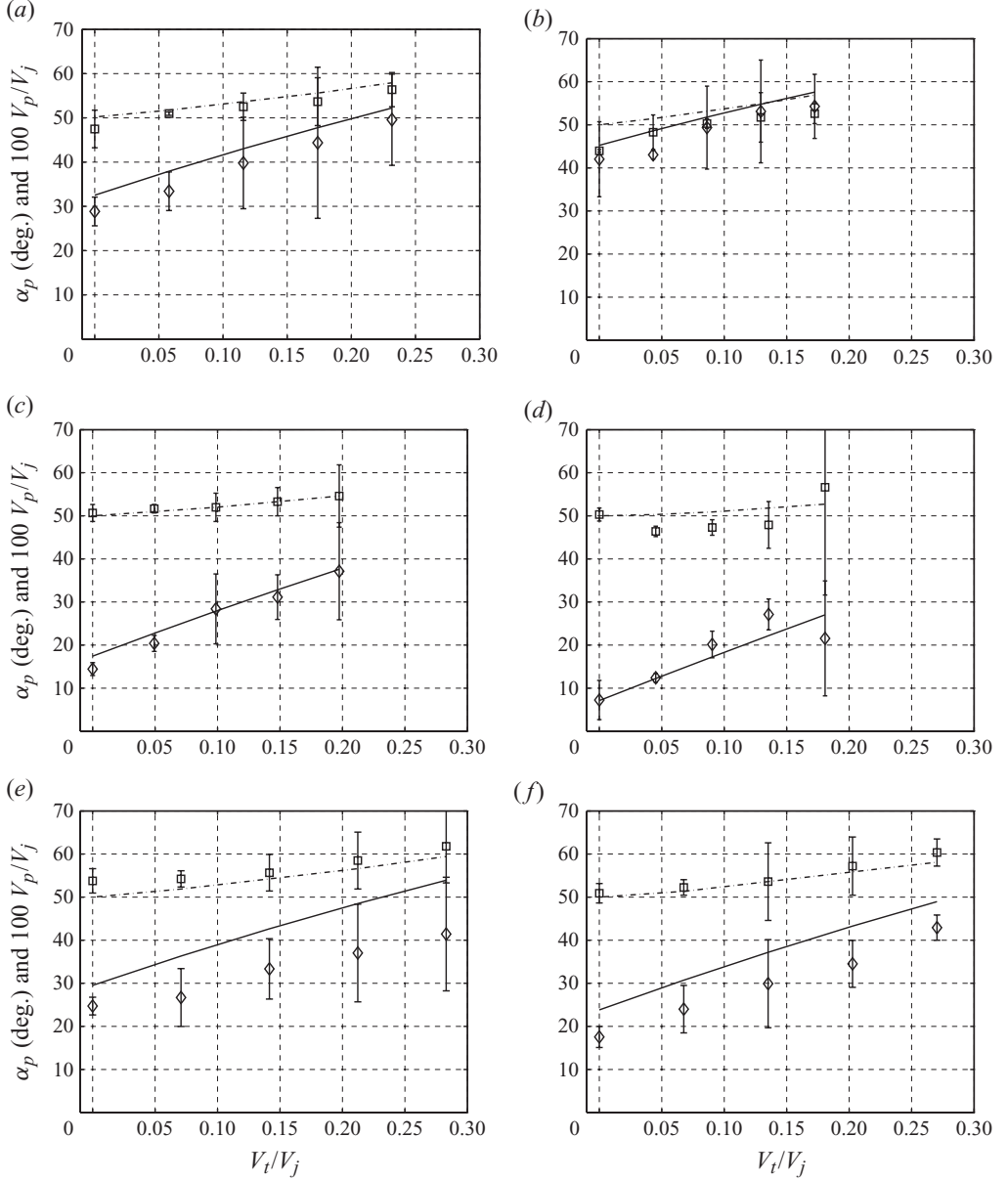


FIGURE 9. Plots of α_p and V_p/V_t times 100 versus translation speed V_t/V_j : $\square$, measured values of V_p/V_j ; $\diamond$, measured values of α_p ; ---, theory for V_p/V_t ; and —, theory for α_p . Error bars for the measurements are shown as vertical lines. Jet conditions I, II, III, IV, V and VI are given in subplots (a), (b), (c), (d), (e) and (f), respectively.

condition. Thus, straight lines were fitted to the position versus time data using the least squares method to obtain V_c and α_c , the magnitude and direction of cavity's leading edge relative to the moving carriage frame of reference. These quantities were converted into a stationary reference frame using the known carriage speed, thus obtaining V_p and α_p . This process was repeated for three different repetitions of the movies for each condition. Plots of α_p and V_p/V_j versus dimensionless carriage translation speed (V_t/V_j) are given in figure 9 for all six jet conditions. The data points

are the averaged measured values and the solid and dash-dotted lines are predictions based on the theory presented in the following. The error bars are the 95 % confidence intervals based on a t -distribution using the three independent samples. It can be seen from the data points in the figures that both V_p/V_j and α_p increase with increasing V_i/V_j for nearly all jet conditions.

In the following, relationships between V_p and α_p and the independent variables of the jet impact are derived and compared to the experimental data in figure 9. Schematics showing the jet and cavity in the reference frame of the water tank and the stagnation point are shown in figures 4(a) and 4(b), respectively. In the reference frame of the jet nozzle, the speed and direction of the jet when it passes the mean water level are V_j and θ_j (relative to vertical), respectively. The speed and direction of the jet when it passes the mean water level in the reference frame of the laboratory are V_i and α_i , respectively, i.e. $\mathbf{V}_i = \mathbf{V}_j + \mathbf{V}_t$. As mentioned above, the velocity of the stagnation point in the laboratory reference frame has magnitude V_p and direction α_p relative to vertical.

3.2.2. Theory

The analysis begins by equating the Bernoulli constant for the stagnation streamlines in the jet and the bulk water in a reference frame fixed with respect to the stagnation point, P , see figure 4. In the laboratory reference frame, the velocity of the stagnation point, $\mathbf{V}_p$, was found to be nearly constant in the above experiments, so we assume that the flow is steady near the jet tip in the stagnation point reference frame. In this reference frame, the flow speed along the stagnation streamline in the tank far upstream is denoted as V_p , and the pressure is $P_{atm} + \rho g z$ (where z points downwards and $z=0$ is the mean water surface in the tank). The Bernoulli equation for the streamline in the tank is

$$\frac{1}{2} V_p^2 + \frac{P_{atm}}{\rho} = \frac{P_p}{\rho} - g z_p, \quad (3.1)$$

where P_p is the pressure at the stagnation point and z_p is the vertical distance from the mean free surface in the tank to the stagnation point. The Bernoulli equation for the stagnation streamline in the jet is

$$\frac{1}{2} |\mathbf{V}_i + \mathbf{V}_p|^2 + \frac{P_{atm}}{\rho} = \frac{P_p}{\rho} - g z_p, \quad (3.2)$$

where the left-hand side is the Bernoulli constant in the jet evaluated at the mean water level. Combining these two equations yields

$$\frac{1}{2} V_p^2 = \frac{1}{2} |\mathbf{V}_i + \mathbf{V}_p|^2. \quad (3.3)$$

Also, $\mathbf{V}_p$ and $\mathbf{V}_i$ can be expressed as

$$\mathbf{V}_p = V_p (\sin(\alpha_p) \mathbf{i} + \cos(\alpha_p) \mathbf{j}), \quad (3.4)$$

$$\mathbf{V}_i = -V_i (\sin(\alpha_i) \mathbf{i} + \cos(\alpha_i) \mathbf{j}). \quad (3.5)$$

Substituting (3.4) and (3.5) into (3.3), an expression for V_p is obtained,

$$V_p = \frac{V_i}{2 \cos(\alpha_p - \alpha_i)}. \quad (3.6)$$

In this expression, both V_p and α_p are unknown, so an additional constraint is needed to close the problem. The additional constraint comes from imposing that

the velocities of the jet water particles are parallel to the jet surface in a reference frame fixed with respect to the stagnation point. This constraint comes from the approximation that (in this reference frame) the jet is nearly steady for the short time between initial impact and pinch-off. In terms of the velocities defined above, this constraint dictates that the components of the velocities V_i and V_p perpendicular to the jet must cancel,

$$V_i \sin(\alpha_i - \theta_j) = V_p \sin(\alpha_p - \theta_j). \quad (3.7)$$

Combining (3.6) with (3.7), it can be shown with some manipulation that

$$\alpha_p = 2\alpha_i - \theta_j, \quad (3.8)$$

which gives α_p as a function of known variables. Using the above expression for α_p in (3.6), V_p can now be expressed in terms of independent variables:

$$V_p = \frac{V_i}{2 \cos(\alpha_i - \theta_j)}. \quad (3.9)$$

The above results also show several interesting things about the flow. Firstly, in the reference frame of the stagnation point, the speed of the incoming flow from the tank is V_p and, by (3.3), the speed of the incoming flow from the jet is V_p as well. Also, from (3.8), the relative angle between the two flows is $\alpha_p - \theta_j = 2(\alpha_i - \theta_j)$. Finally, when the jet is stationary, $\alpha_i = \theta_j$ and $V_p = V_i/2$, as noted by others (see, for example Birkhoff & Zarantonello 1957).

Expressions (3.8) and (3.9), which give the magnitude and the direction of the underwater jet tip velocity (V_p and α_p) as a function of known jet parameters (V_i , α_i and θ_j) are plotted along with the experimental data in figure 9. As can be seen in the plots, the model is a good approximation to the data. The model presents the largest deviations from the measured values in the curves of α_p versus V_i/V_j for conditions V and VI, subplots (e) and (f), respectively. These discrepancies may be associated with the fact that these two conditions have the smallest values of Fr_j , and it is possible that the underwater jet trajectory may be more affected by the gravitational effects associated with the motion of the free surface of the pool than in the other cases. The effect of the free surface of the pool is not included in the model.

Finally, an interesting observation can be made from the fact that a steady state can only be obtained if the stagnation point moves downwards and upstream, i.e. $\alpha_p < \pi/2$. This condition is equivalent to (see (3.8)),

$$\alpha_i < \frac{\pi}{4} + \frac{\theta_j}{2}. \quad (3.10)$$

Now, since

$$\mathbf{V}_i = \mathbf{V}_j + \mathbf{V}_t, \quad (3.11)$$

it can easily be shown that

$$\tan \alpha_i = \frac{V_j \sin \theta_j + V_t}{V_j \cos \theta_j}, \quad (3.12)$$

which when combined with (3.10), yields

$$\frac{V_t + V_j \sin \theta_j}{V_j \cos \theta_j} < \tan \left(\frac{\pi}{4} + \frac{\theta_j}{2} \right) = \frac{\cos \theta_j}{1 - \sin \theta_j}. \quad (3.13)$$

This reduces to

$$V_t < V_j, \quad (3.14)$$

which is satisfied for all the conditions presented in this work. This equation implies that if the relative translational velocity of the jet nozzle exceeds the jet speed, it is impossible for the steady-state conditions to be established for the given boundary conditions. Unfortunately, with the present experimental setup it was not possible to achieve sufficiently high carriage speeds or sufficiently low jet speeds to test this result.

3.3. Temporal evolution of the underwater craters – pinch-off depth and time

3.3.1. Measurements

The high-speed movies were analyzed to determine the shape of the upstream and downstream craters versus time and the depth and time of the pinch-off of the forward crater for all experimental conditions.

To obtain this data, the shapes of the craters were determined by selecting about 50 points on the crater walls from each movie frame. The underwater jet edges were also digitized, when visible; however, the jet edge results should be regarded as qualitative since it is typically difficult to identify the exact position of the jet edges on the movie frames. Sample plots of the cavity contours for jet conditions I and V are shown in figures 10 and 11, respectively. In each plot, contours are given for translation speeds of $V_t = 0.0, 0.30$ and 0.60 m s^{-1} and for each translation speed at four equally spaced times ($1/4t_{po}$, $1/2t_{po}$, $3/4t_{po}$ and t_{po}), where t_{po} is the measured pinch-off time of the forward crater. The results are presented in a reference frame that is translating with the jet nozzle.

The pinch-off depth normalized by T_j is plotted versus V_t/V_j for each jet condition in figure 12(a–f) and the pinch-off time normalized by the jet time scale, T_j/V_j , is plotted versus V_t/V_j in figure 13(a–f). With the exception of jet condition II (subplots (b) in figures 12 and 13) both the pinch-off depths and pinch-off times decrease monotonically with translation speed in each jet condition. Jet condition II has the largest angle $\theta_j = 45.2^\circ$ relative to vertical (the next highest angle is for jet condition I where $\theta_j = 32.5^\circ$). It is speculated that, due to this large angle, the crater development might be more influenced by the impact-induced motions of the free surface of the tank water than in the other cases.

3.3.2. Theory

In the following, a simple model is developed for the prediction the crater shapes and the pinch-off depths and times. In this model, a crater shape is first computed with a theory that does not include gravity. Then, the effect of gravity is added to determine the time history of the crater shape up to pinch-off.

The shape of the cavity in the absence of gravity can be approximated from the problem of the steady impact of two jets as presented in Milne-Thomson (1968, pp. 299). A schematic of this jet impact problem is shown in figure 14 in the reference frame of the stagnation point. In the analysis, the incoming jet A_1 (in our case the jet from the nozzle) has speed U , thickness h_1 and is directed along the x axis. As is shown in Milne-Thomson (1968), in the reference frame of the stagnation point, the other incoming jet and both outgoing jets also have speed U . The second incoming jet A_2 (in our case the tank water) has thickness h_2 and velocity direction angle α relative to the direction of the velocity of jet A_1 . Thus, in the present case $U = V_p$ and $\pi - \alpha = \alpha_{rel} = \alpha_p - \theta_j = 2(\alpha_i - \theta_j)$. The solution to this problem is found by solving the case of two impacting planar jets of finite thickness (h_1 and h_2) and then extrapolating the solution to the asymptotic case where the thickness of the jet comprised of the

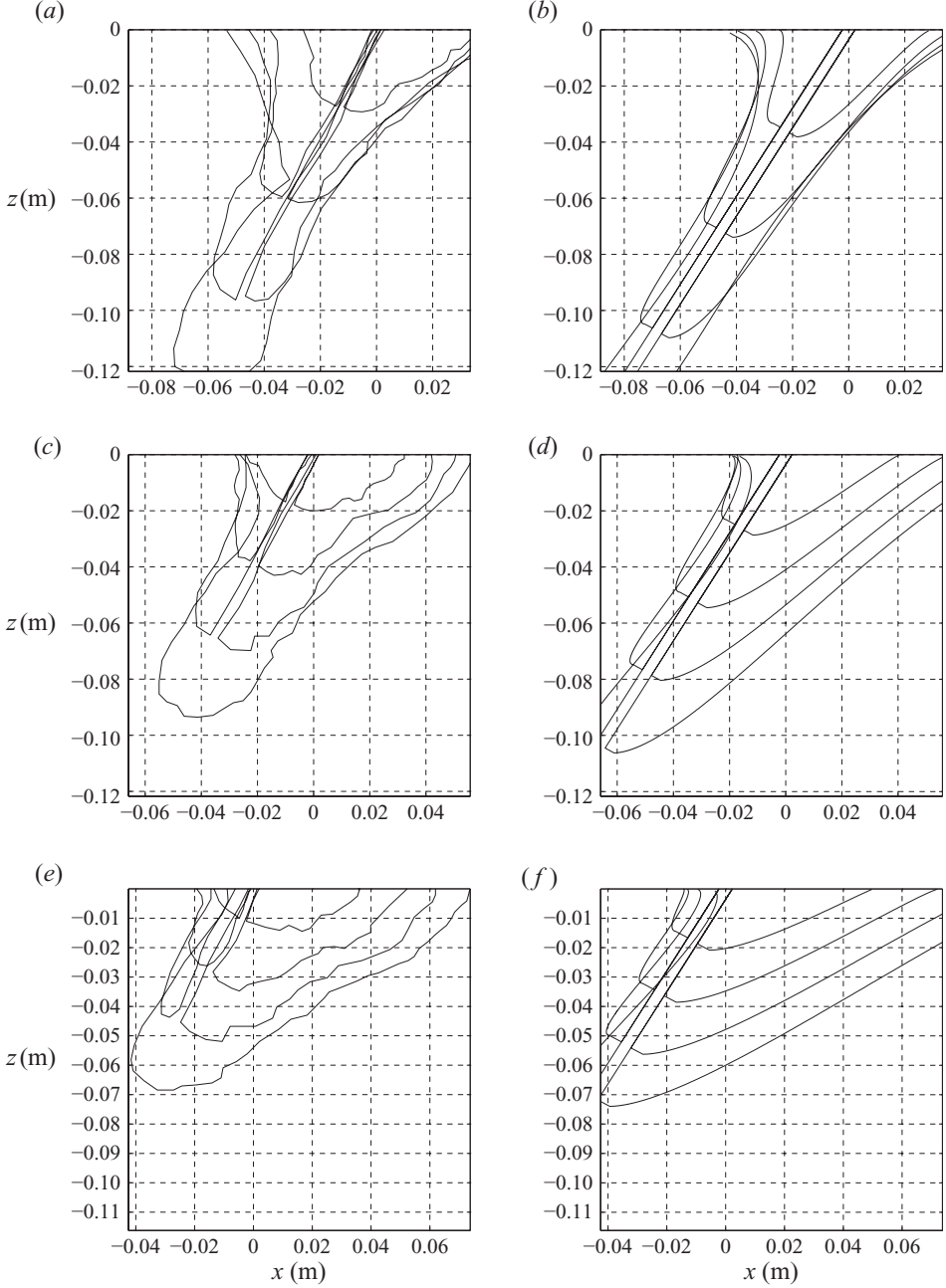


FIGURE 10. Crater temporal evolution for jet condition I ($Fr_j = 13.6$, $\theta_j = 32.5^\circ$). Measured contours are given in the left column for $V_t = 0.00 \text{ m s}^{-1}$ (a), $V_t = 0.30 \text{ m s}^{-1}$ (c) and $V_t = 0.60 \text{ m s}^{-1}$ (e). The contours are given in a reference frame moving with the jet nozzle and at four times $1/4t_{po}$, $1/2t_{po}$, $3/4t_{po}$ and t_{po} , where t_{po} is the time of pinch-off of the forward cavity. The three sets of contours in the right column were computed with the theoretical model for the same jet conditions and times as the corresponding experimental data.

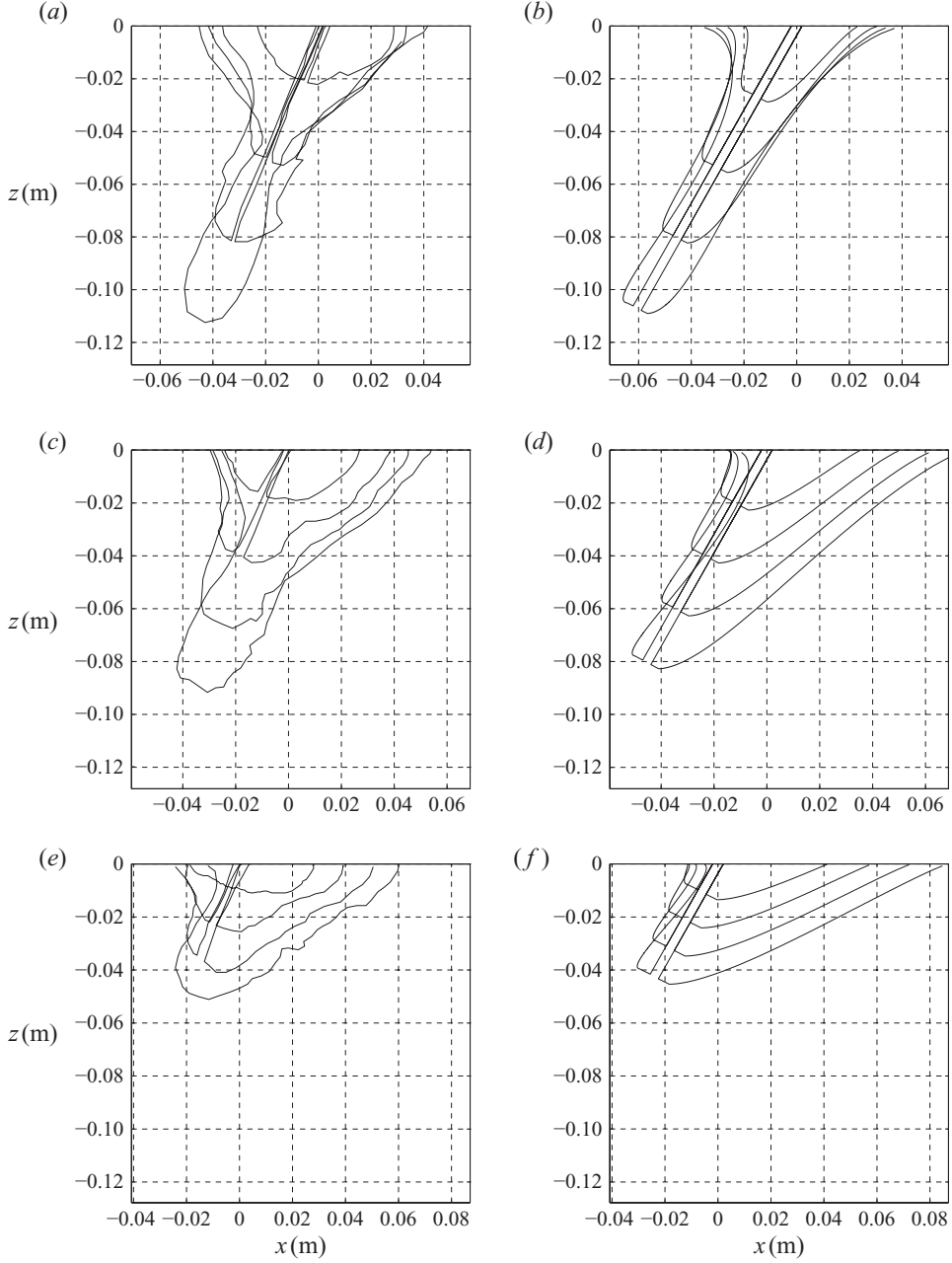


FIGURE 11. Crater temporal evolution for jet condition V ($Fr_j = 11.4$, $\theta_j = 29.5^\circ$). Measured contours are given in the left column for $V_t = 0.00 \text{ m s}^{-1}$ (a), $V_t = 0.30 \text{ m s}^{-1}$ (c) and $V_t = 0.60 \text{ m s}^{-1}$ (e). The contours are given in a reference frame moving with the jet nozzle and at four times $1/4t_{po}$, $1/2t_{po}$, $3/4t_{po}$ and t_{po} , where t_{po} is the time of pinch-off of the forward cavity. The three sets of contours in the right column were computed with the theoretical model for the same jet conditions and times as the corresponding experimental data.

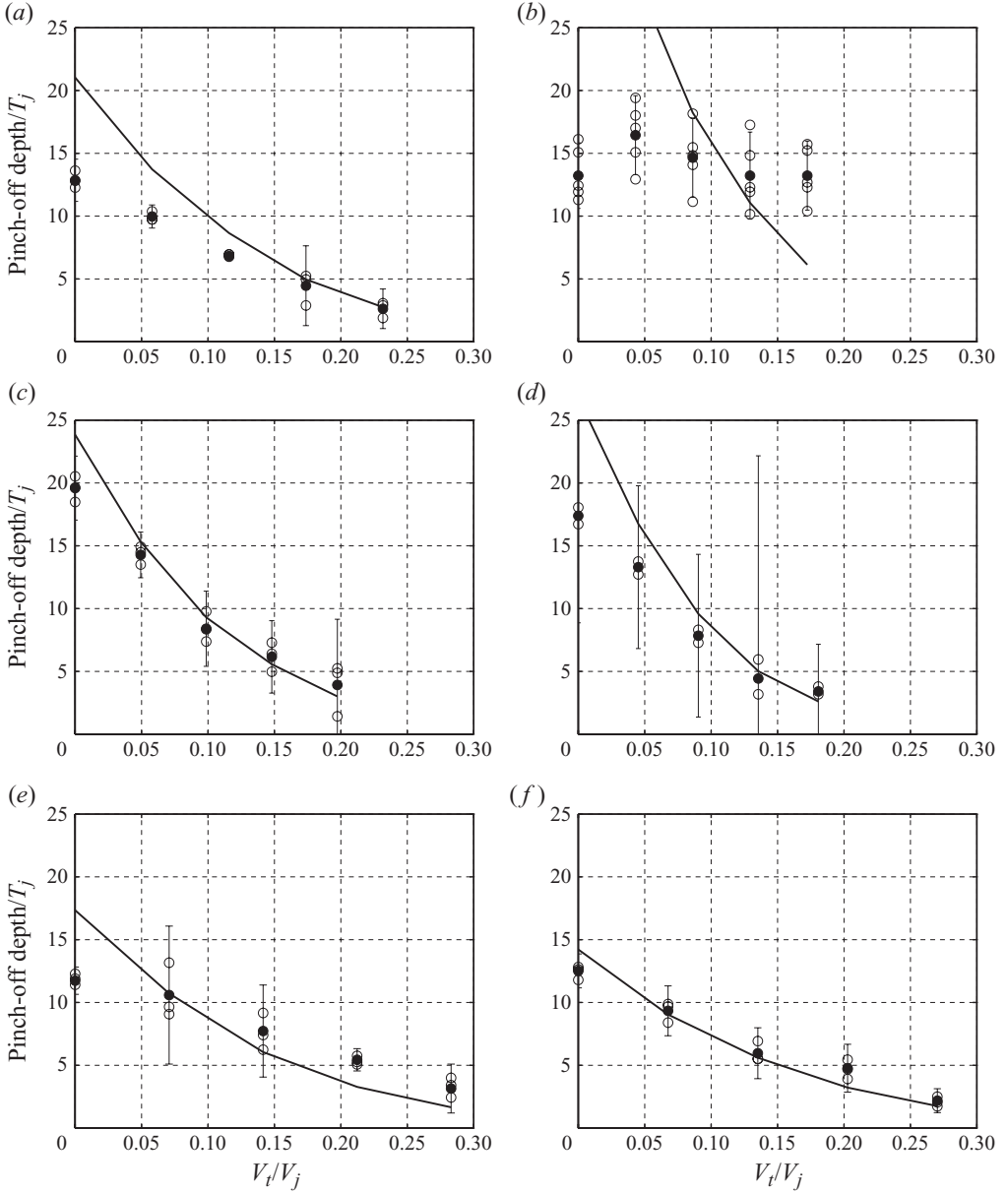


FIGURE 12. Pinch-off depth versus V_t/V_j . Jet conditions I, II, III, IV, V and VI in graphs (a), (b), (c), (d), (e) and (f), respectively. The open circles are individual data points and the filled circles are average values. The vertical lines are error bars. The solid lines are predictions from the theory.

tank water, h_2 , tends to infinity (see figure 14). The two outgoing jets, B_1 and B_2 , have thicknesses k_1 and k_2 and directions β and γ relative to the direction of the velocity of stream A_1 , respectively. The conservation of momentum in the two coordinate directions yield

$$h_1 + h_2 \cos \alpha - k_1 \cos \beta - k_2 \cos \gamma = 0, \quad (3.15)$$

$$h_2 \sin \alpha - k_1 \sin \beta - k_2 \sin \gamma = 0. \quad (3.16)$$

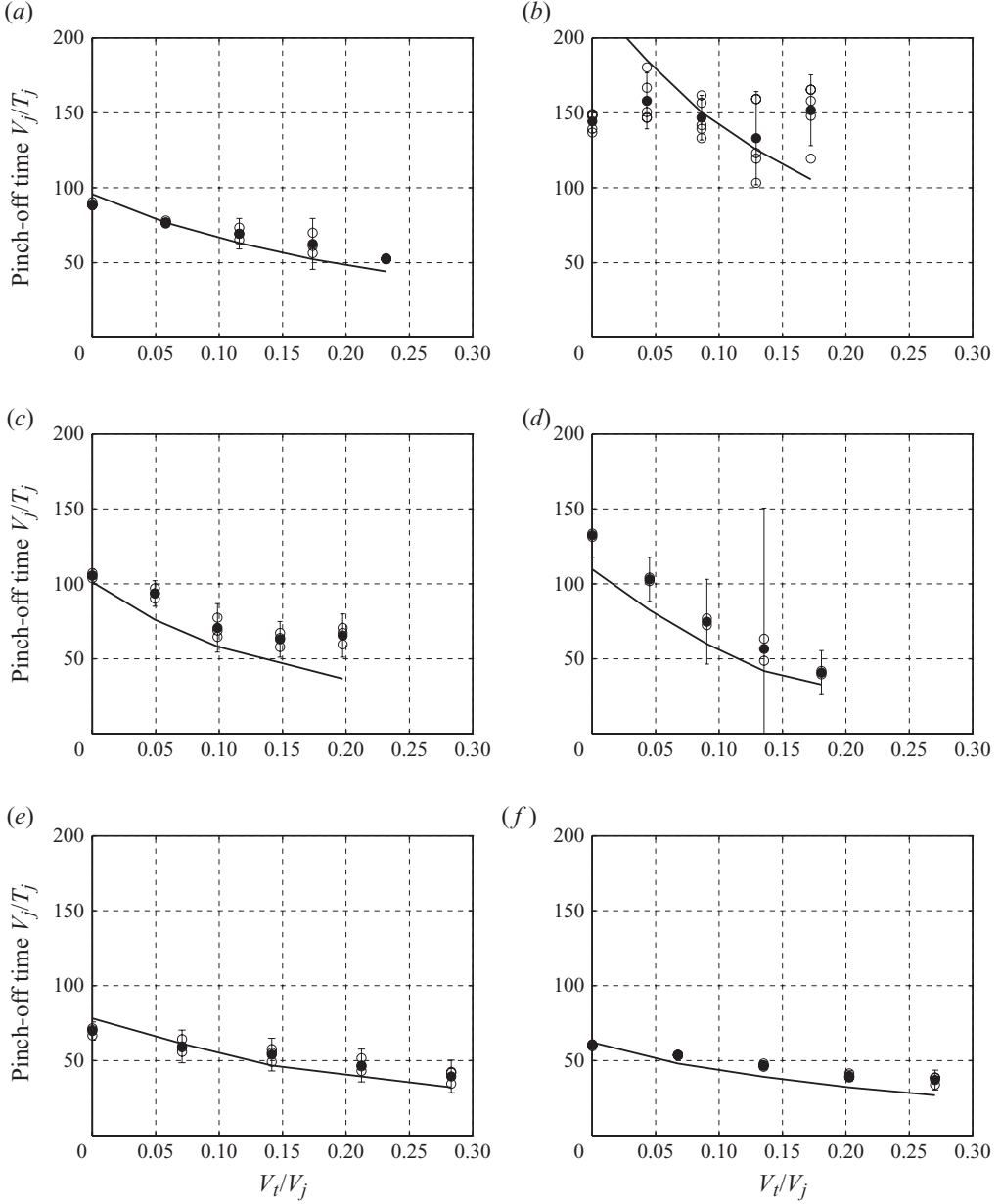


FIGURE 13. Pinch-off time versus V_t/V_j . Jet conditions I, II, III, IV, V and VI in graphs (a), (b), (c), (d), (e) and (f), respectively. The open circles are individual data points and the filled circles are average values. The vertical lines are error bars. The solid lines are predictions from the theory.

The conservation of mass gives the constraint,

$$h_1 + h_2 = k_1 + k_2. \quad (3.17)$$

The x and y coordinates of the shape of the free surfaces of the jet impact are

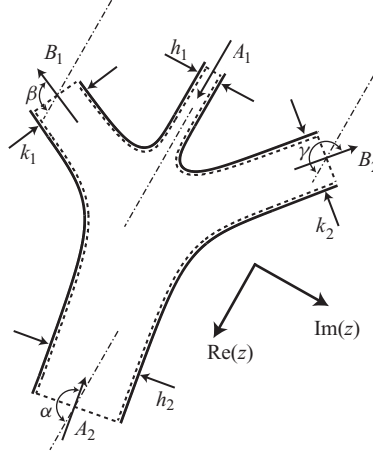


FIGURE 14. Definition of symbols and geometry for the problem of two steady jets of finite thickness colliding with one another. In the present case, the jet A_1 is the plunging jet and A_2 is the water in the tank. The thicknesses of the outgoing jets are taken as equal, $k_1 = k_2$. The flow is shown in the reference frame of the stagnation point. The depicted shape of the jet impact region is a solution of (3.15)–(3.18) with $\alpha = 170^\circ$. The variable $z = x + iy$ is the complex position vector and the dash-dotted lines are parallel to the x axis, which is parallel to the direction of the plunging jet.

given by

$$\begin{aligned} \pi z = & \frac{i}{2} \left(-h_2 \alpha e^{-i\alpha} + k_1 \beta e^{-i\beta} + k_2 \gamma e^{-i\gamma} \right) \\ & + h_1 \log \left(\sin \frac{\theta}{2} \right) + h_2 e^{-i\alpha} \log \left(\sin \frac{\theta + \alpha}{2} \right) \\ & - k_1 e^{-i\beta} \log \left(\sin \frac{\theta + \beta}{2} \right) - k_2 e^{-i\gamma} \log \left(\sin \frac{\theta + \gamma}{2} \right), \end{aligned} \quad (3.18)$$

where $z = x + iy$, x and y are the Cartesian coordinates in the plane of the flow, $i = \sqrt{-1}$ and θ is an independent parameter that varies from 0 to 2π .

Given the values of h_1 , h_2 and α , for the incoming jets, there are only three equations, (3.15), (3.16) and (3.17), for the four unknowns k_1 , k_2 , β and γ . Thus, the system is indeterminate. However, in the present case because the second incoming jet consists of the water in the tank, we have $h_2 \gg h_1$. It is found that in this case $\beta \rightarrow \gamma \rightarrow \alpha$ as long as k_2/k_1 is not too large or too small, i.e. as long as the stagnation point is not too close to the surfaces of the large jet, a condition which is satisfied in the present experiment. The shapes of the free streamlines were computed numerically for fixed values of $h_1 = T_j$ and increasing values of h_2 until enough convergence to the asymptotic solution was achieved. The specific procedure used to check for the convergence of the solution is described in Gomez-Ledesma (2004). Sample cavity shapes computed for $\theta_j = 20^\circ$ and $V_i = 0.0$ and $0.1V_j$ are shown in figure 15.

The effect of gravity on the above-derived non-gravitational cavity shape is now considered. Figure 16(a) shows a schematic of the underwater plunging jet. The velocity and direction of penetration of the jet tip in a reference frame fixed with respect to the receiving pool (V_p and α_p) are given by expressions (3.6) and (3.8). The tip region of the jet and craters is shown in figure 16(b). The solid line represents the shape of the underwater craters as predicted by the non-gravitational theory, while

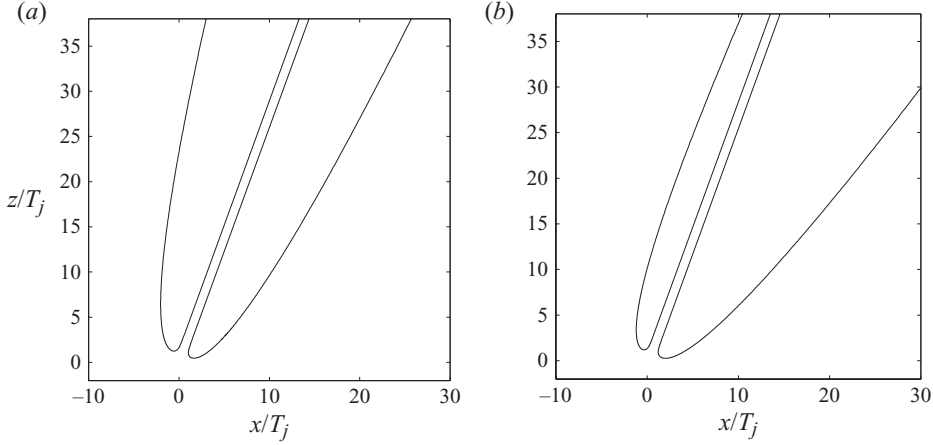


FIGURE 15. Shape of the underwater craters computed using the steady-state zero-gravity theory due to Schwarz (Milne-Thomson 1968). $\theta_j = 20^\circ$, $V_i/V_j = 0.0$ (a). $V_i/V_j = 0.1$ (b). The jet is translating from right to left in (b).

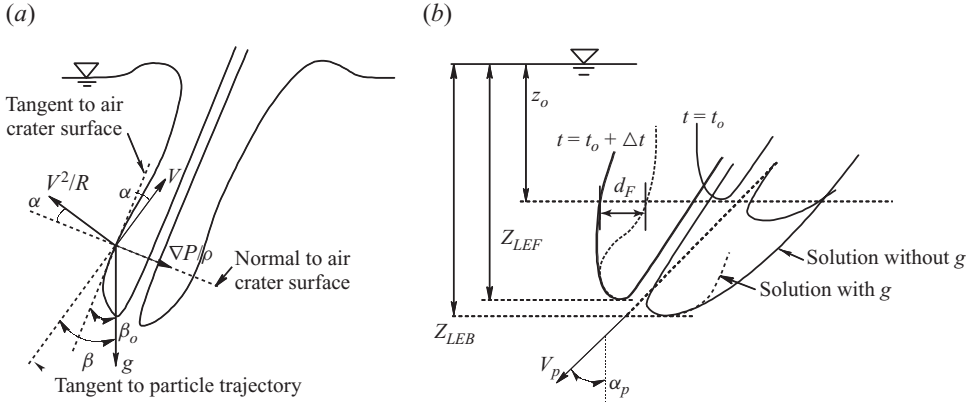


FIGURE 16. Schematic of the jet and craters showing the definition of symbols for the theory predicting the effect of gravity on the crater shape.

the dashed line depicts the solution under the influence of gravity. Expressions (3.6) and (3.8) are valid for the non-gravitational solutions and approximately valid for the case with gravity; therefore, for both solutions the jet leading edge is expected to be at the same location at a given time. A good first approximation to the deformation of the crater under the effects of gravity can be achieved by only accounting for the effects of the gravity-induced deformation velocity perpendicular to the crater wall (v'). Let $d_F(z_o, t)$ be the difference in the horizontal position of the front crater predicted by the gravitational and non-gravitational theories at a given depth z_o , as shown in figure 16(b). Also, let z_{LEF} be the depth of the leading edge of the front crater at time t and t_o be the time at which the leading edge of the front cavity was at $z_o < z_{LEF}$. The displacement d_F can be estimated by integrating the horizontal projections of all the infinitesimal displacements seen by the particles at height z_o on

the front wall of the crater between the times t_o and t :

$$d_F(z_o, t) = \int_{t_o}^t v'(z_o, t) \cos \beta_o^F(z_o, t) dt, \quad (3.19)$$

where β_o^F is the angle between the tangent line to the front crater and the vertical at z_o , see figure 16(a).

Following the work of Oguz *et al.* (1995), we assume that on the outer cavity walls, away from the jet tip, the primary force inducing the inward motion of the cavity is due to the difference between the hydrostatic pressure far from the cavity at the depth of any point of interest and the nearly atmospheric pressure inside the cavity. Thus, we assume

$$v' = \lambda \sqrt{gz}, \quad (3.20)$$

where λ is an empirical constant. Of course, (3.20) is only a scaling argument; in reality, the crater wall velocity will vary with time. Using this expression for v' , (3.19) can be written as

$$d_F(z_o, t) \approx \lambda \sqrt{gz_o} \int_{t_o}^t \cos \beta_o^F(z_o, t) dt. \quad (3.21)$$

If β_o^F is approximated by the values of the non-gravitational solution at each time and realizing that $dt = dz/(V_p \cos \alpha_p)$, this expression can be transformed into an integral over distance

$$d_F(z_o, t) \approx \frac{\lambda \sqrt{gz_o}}{V_p \cos \alpha_p} \int_{z_o}^{z_{FLE(t)}} \cos \beta_o^F(z) dz. \quad (3.22)$$

Similarly, for the back crater,

$$d_B(z_o, t) \approx \frac{\lambda \sqrt{gz_o}}{V_p \cos \alpha_p} \int_{z_o}^{z_{BLE(t)}} \cos \beta_o^B(z) dz. \quad (3.23)$$

In implementing these latter equations, the shape of the underwater craters was computed in time increments of 4 ms until the minimum distance between the outer walls of the air craters and the jet centreline was less than or equal to $0.5T_j$, defined as pinch-off. A value of $\lambda = 0.56$ was determined by best fit of the computed pinch-off times and depths to the data obtained from the experiments. This value was found by minimizing the residual

$$R(\lambda) = \sqrt{\frac{\sum [1 - \tau_{po,i}^{theo}(\lambda)/\tau_{po,i}^{exp}]^2 + \sum [1 - \delta_{po,i}^{theo}(\lambda)/\delta_{po,i}^{exp}]^2}{n}}, \quad (3.24)$$

where $\tau_{po,i}^{exp}$ and $\delta_{po,i}^{exp}$ are the non-dimensional pinch-off time and depth for a given jet condition and translation velocity and $\tau_{po,i}^{theo}(\lambda)$ and $\delta_{po,i}^{theo}(\lambda)$ are their equivalent theoretical predictions for a given value of λ . The sums are over all the translation velocities and jet conditions except for jet condition II (which is out of the range of validity of this model as explained later).

Sample predictions of crater profiles are plotted next to the corresponding measured crater profiles in figures 10 and 11. As can be seen from the plots, the shapes of the underwater craters computed with the model are in good qualitative agreement with the measured crater shapes, replicating the general trends of the crater wall behaviour. Perhaps the largest difference between the two sets of profiles is that the model predicts a somewhat uniform collapse along the crater walls, leading to a continuous reduction with time of crater's width at all depths, while in the measured

profiles, the gravity-induced deformations are accentuated around the region where the pinch-off takes place. This is likely because the effect of streamline curvature on v' , which is strongest at the leading edge of the cavity, is not considered in the theory.

The theoretical values of the pinch-off depths and times are compared to the experimental data in figures 12 and 13, respectively. For jet conditions I, III, IV, V and VI, the model provides fairly accurate predictions of values and trends of the experimental data. For condition II, the model is not in good agreement with the observations, as can be seen in the plots in figures 12(b) and 13(b). The jet inclination θ_j for conditions I, III, IV, V and VI ranges from 7.1 to 32.5° , while the jet inclination for condition II is considerably larger, $\theta_j = 45.2^\circ$. It is speculated that for large angles of inclination, the collapse process may be dominated by the dynamics of the pressure field of the nearby upper free-surface splash, voiding the basic assumptions of this model. In addition, these conditions may be sufficiently different such that the simplistic scaling given by (3.20) cannot accommodate a correction across this wider range of behaviour.

4. Conclusion

The impact of a two-dimensional jet that is translating over a liquid pool is explored with experiments and theory. In all the cases studied herein, the jet is tilted in the direction of translation and it penetrates the pool creating upstream and downstream craters and splashes. After impact, the jet tip quickly slows down and then maintains a nearly constant velocity downwards and in the direction of translation. This velocity is predictable with a relatively simple steady-state theory. The temporally evolving shapes of the upstream and downstream craters are strongly affected by the jet translation speed. In all the cases, air is first entrained into the pool when the forward crater pinches off at a depth between the mean water level and the jet tip. The depth and time of this pinch off decrease as the translation speed increases. This trend indicates that while the forward translation speed of the jet may tend to increase the forward motion of the front crater wall, this effect is over powered by the effect of the jet translation narrowing the distance between the front crater wall and the jet as time increases. The cavity shapes and pinch-off depths and times are predicted by a theory based on an initial cavity shape from steady theory without gravity and a gravity-induced inward velocity that increases as the square root of the depth. The experimental and theoretical cavity shapes match well as long as the impact angle of the jet relative to vertical is not too large. Though the jet penetrated into the pool in all the cases studied herein, it is speculated that there may be conditions, perhaps larger jet angles relative to vertical and higher translation speeds for which the jet essentially bounces off of the free surface of the pool, creating a large upstream splash zone.

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